Muhammad **Ayyan**

AI/ML Engineer . Data Science

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Summary

Data Science AI/ML Professional skilled in data preprocessing, SQL extraction and building supervised ML models (Scikit-Learn, XGBoost) for actionable insights. Proficient in Python, Pandas, NumPy, Data visualization, and Pipeline automation.Experienced in model deployment and MLOps practices using Docker, Kubernetes, Git and MLflow, with strong analytical communication skills to collaborate effectively with product and engineering teams.

Tools & Technologies

- **Languages & Libraries:**
  Python – SQL - Langchain - LangGraph - Tensorflow - Scikit-learn – NumPy – Pandas – Matplotlib – FastAPI - Pydantic - Hugging Face - ML Flow - DVC - Optuna - Statistics
- **AI & Machine Learning:**
  Agents - RAG - LLMs - Fine-Tuning - Transformer - NLP - ANNs - CNNs - RNNs - LSTM - Supervised & Unsupervised Learning - Hyperparameter Tuning – Model Evaluation – EDA – Feature Engineering - MLFlow - Docker - Kubernetes - CI/CD - EC2 - S3
- **Tools & Platforms:**
  VSCode - Jupyter Notebook - Git - GitHub - MySQL - Ollama - Excel - PowerBI

Experience

NETSOL TECHNOLOGIES Inc. || AI/ML Trainee

Onsite

◇ Engineered and deployed advanced machine learning model workflows, seamlessly bridging data preprocessing, data visualization, feature engineering, and containerized model monitoring.

◇ Cleaned and analyzed complex datasets using Python, Pandas, NumPy and SQL.Built supervised models with interpretability checks, packaged the data pipelines using Docker,and simulated production environments on local Kubernetes (Minikube) clusters with Grafana/Prometheus monitoring.

◇ Delivered reproducible data pipelines Jupyter notebooks and concise visualizations with Matplotlib and Seaborn to communicate analytical insights. Maintained clean,version-controlled (Git) code, demonstrating full-lifecycle capabilities from raw data extraction to scalable deployment. .

OTHER PROJECTS

Vehicle Insurance Prediction |GitHub | Demo

◇ Built an end-to-end MLOps pipeline for vehicle insurance data with MongoDB Atlas ingestion, logging, and exception handling.

◇ Developed YAML-based data validation, transformation, and feature engineering modules.

◇ Deployed real-time ML inference using FastAPI and a web-based UI.

Tools : Python,Transformers,PyTorch,FastAPI,Skit-Learn,MongoDB,Hugging Face, Pandas, FastAPI, Docker, Git,Dagshub.

- **Kidney Disease Clasification |GitHub | Demo**

◇ Built a CNN-based deep learning pipeline for kidney disease classification.

◇ Implemented MLOps with MLflow, DVC, and Flask for tracking and data versioning.

Tools : Python, Scikit-learn, Pandas, Docker,TensorFlow,Keras,ML-Flow,DVC,PythonHugging Face Spaces

Certifications

Udemy: Machine & Deep Learning With Python
View Certificate

Udemy: Python FOR Machine Learning
View Certificate

Academics

Virtual University of Pakistan *Lahore, Pakistan*
B.S. in Computer Science